TCP/IP

- Ch-4

Internetworking

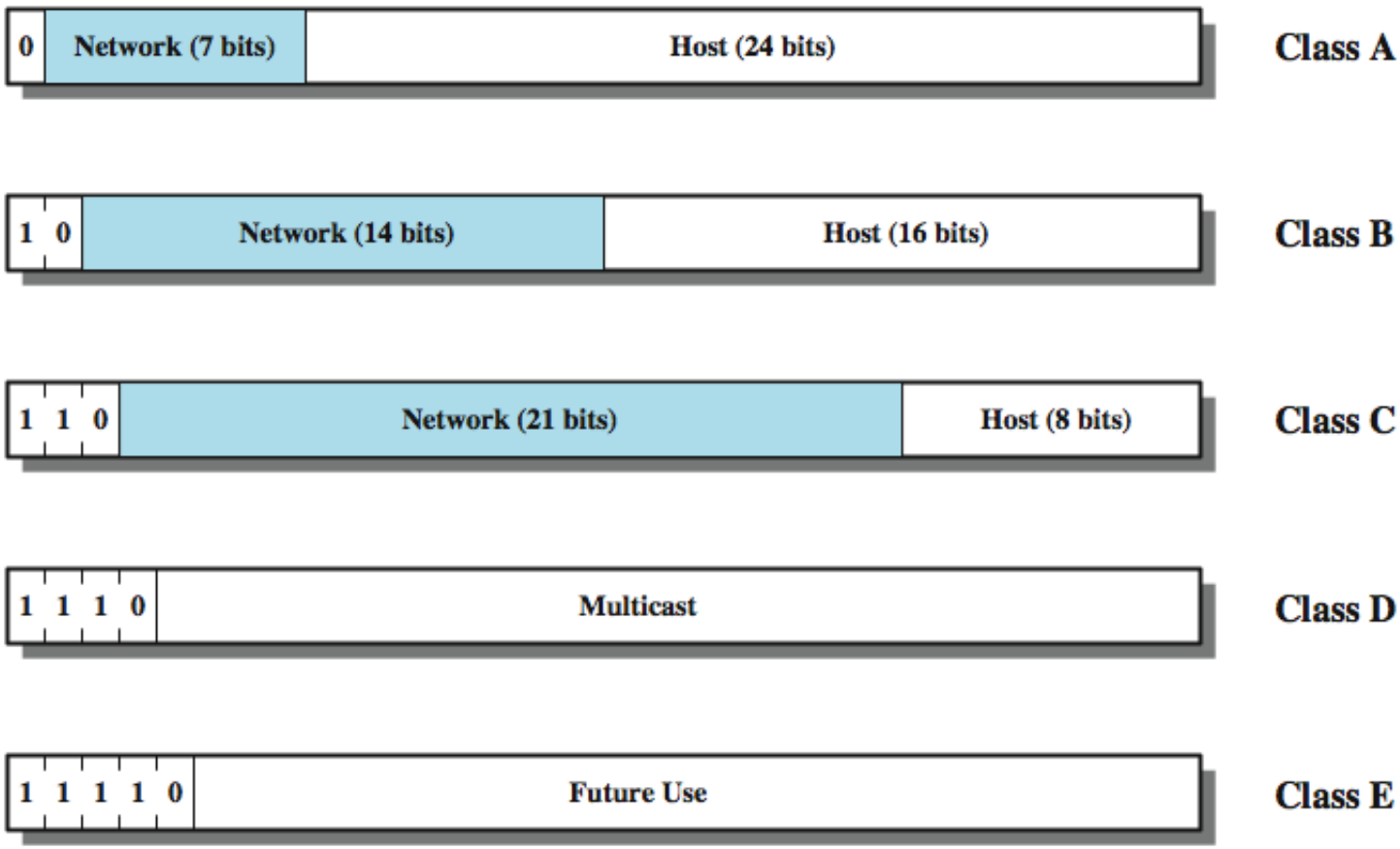
Ch-4
Internet Protocol (IP) v4

- IP version 4
- defined in RFC 791
- part of TCP/IP suite
- two parts
  - specification of interface with a higher layer
    - e.g. TCP
  - specification of actual protocol format and mechanisms
- will (eventually) be replaced by IPv6

IP Address classes

Address Class	Network Bytes	Host Bytes	Beginning Bit Values	Addresses (First Octet)
A	1 (8 bits)	3 (24 bits)	0	1–126; 0 and 127 are reserved
B	2 (16 bits)	2 (16 bits)	10	128–191
C	3 (24 bits)	1 (8 bits)	110	192–223
D	N/A	N/A	1110	224–239
E	N/A	N/A	11110	240–254; 255 is reserved

IPv4 Address Formats



IP Addresses - Class A

- start with binary 0
- all 0 reserved
- 01111111 (127) reserved for loopback
- range 1.x.x.x to 126.x.x.x
- 2^8 = 256 network address
- 2^{24} -2= 16,777,214 host addresses
- all allocated

IP Addresses - Class B

- start with binary 10
- range 128.x.x.x to 191.x.x.x
- second octet also included in network address
- 2^{16} = 65,536 class B addresses
- 2^{16} -2= 65,534 host address
- all allocated

IP Addresses - Class C

- start with binary 110
- range 192.x.x.x to 223.x.x.x
- second and third octet also part of network address
- 2^{24} = 16,777,216 addresses for network.
- nearly all allocated
  - 2^{8} -2 =254 host address

Subnets and Subnet Masks

- allows arbitrary complexity of internetworked LANs within organization
- insulate overall internet from growth of network numbers and routing complexity
- site looks to rest of internet like single network
- each LAN assigned subnet number
- host portion of address partitioned into subnet number and host number
- local routers route within subnetted network
- subnet mask indicates which bits are subnet number and which are host number

Subnet Mask Calculation

	Binary Representation	Dotted Decimal
IP address	11000000.11100100.00010001 .00111001	192.228.17.57
Subnet mask	11111111.11111111.11111111 .11100000	255.255.255.224
Bitwise AND of address and mask (resultant network/subnet number)	11000000.11100100.00010001 .00100000	192.228.17.32
Subnet number	11000000.11100100.00010001 .001	1
Host number	00000000.00000000.00000000 .00011001	25

IP Versions

- IP v 1-3 defined and replaced
- IP v4 - current version
- IP v5 - streams protocol
- IP v6 - replacement for IP v4
  - during development it was called IPng (IP Next Generation)

Why Change IP?

- Address space exhaustion
  - two level addressing (network and host) wastes space
  - network addresses used even if not connected
  - growth of networks and the Internet
  - extended use of TCP/IP
  - single address per host
- requirements for new types of service

IPv6 RFCs

- RFC 1752 - Recommendations for the IP Next Generation Protocol
  - requirements
  - PDU formats
  - addressing, routing security issues
- RFC 2460 - overall specification
- RFC 2373 - addressing structure
- many others

IPv6 Enhancements

- expanded 128 bit address space
- improved option mechanism
  - must not examined by intermediate routes
- dynamic address assignment
- increased addressing flexibility
  - any cast & multicast
- support for resource allocation
  - labeled packet flows

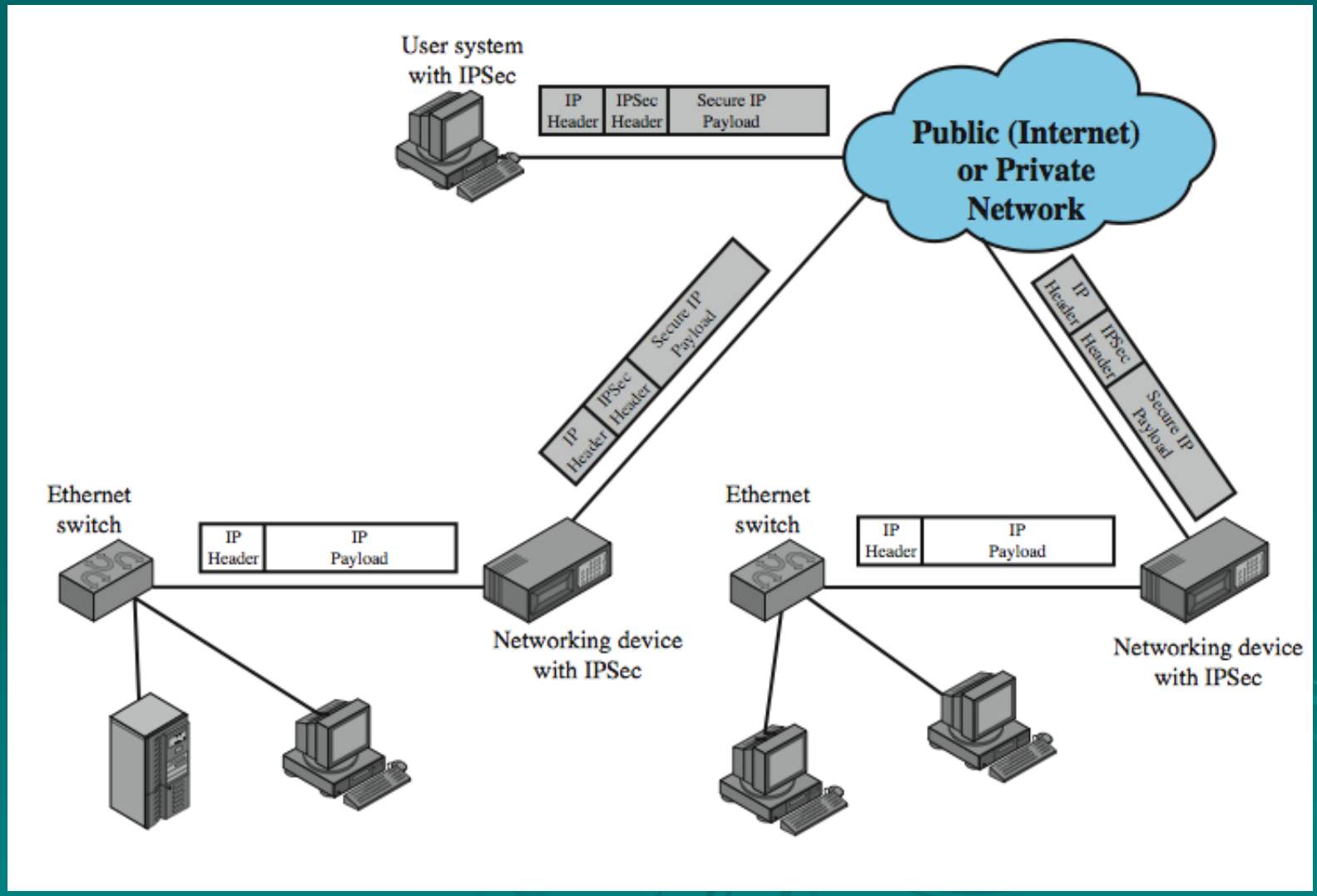
Virtual Private Networks

- set of computers interconnected using an insecure network
  - e.g. linking corporate LANs over Internet
- using encryption & special protocols to provide security
  - to stop eavesdropping & unauthorized users
- proprietary solutions are problematical
- hence development of IPSec standard

IPSec

- RFC 1636 (1994) identified security need
- encryption & authentication to be IPv6
- but designed also for use with current IPv4
- applications needing security include:
  - branch office connectivity
  - remote access over Internet
  - extranet & intranet connectivity for partners
  - electronic commerce security

IPSec Scenario



IPSec Benefits

- provides strong security for external traffic
- resistant to bypass
- below transport layer hence transparent to applications
- can be transparent to end users
- can provide security for individual users if needed

IPSec Functions

- Authentication Header
  - for authentication only
- Encapsulating Security Payload (ESP)
  - for combined authentication/encryption
- a key exchange function
  - manual or automated
- VPNs usually need combined function

IP Address Planning

1. Determine network and host requirements.
2. Satisfy host and network requirements.
3. Determine the subnet mask.
4. Determine the network addresses.
5. Determine the directed broadcast addresses for your networks.
6. Determine the host addresses for your networks.

Step 1: Determine Network and Host Requirements

In this step, you need to do two things:

- Determine the number of hosts that do, or will, exist on the largest segment in your network.
- Determine the maximum number of segments that you have in your network—this will tell you how many networks, or subnets, you'll need.

Step 2: Satisfy Host and Network Requirements

In the second step, you'll use three formulas:

- 2^S >= number of networks you need (S represents subnet bits)
- $2^H - 2$ >= number of hosts on your largest segment (H represents host bits)
- $S + H$ <= total number of host bits you have for a class of address

Step 3: Determine the Subnet Mask

- Now that the hardest part is over, the remaining four tasks are easy. At this point, you know the number of subnet bits you need. However, when dealing with networking and subnet masks, a subnet mask's network portion contains both network *and* subnet bits. Remember the default number of networking bits for a class address: A is 8, B is 16, and C is 24.

Step 4: Determine the Network Addresses

- In step 4, you need to figure out the networks that you created with your new subnet mask. Since IP addressing is done in binary, network addresses will always increment in a multiple of something.
- All host bits become 0

Step 5: Determine the Directed Broadcast Addresses

- After figuring out all of your subnets, you next need to determine the directed broadcast address for each subnet. This is very simple.
- The directed broadcast of a subnet is *one number less* than the next network number.
- Also, the broadcast address has all of its hosts bits (in the subnet) set to binary 1s.

Step 6: Determine the Host Addresses

- Step 6 is the easiest step. Recall that any address between the network and directed broadcast address is a host address for a given network.

Example

No	Se14gment	No- of user (hosts)
1	Employee	14
2	Guest 1	8
3	Guest 2	8
4	Guest 3	8
5	Guest 4	8
6	Guest 5	8
7	Management 1	4
8	Management 2	4
9	Management 3	4
10	Management 4	4
11	ICT 1	2
12	ICT2	2
13 14	R1 ISPN	2 2

Step-2

1. $2^S >= 14$ subnets; in this example, $S = 4$, which would result in 16 subnets.
2. $2^H - 2 >= 14$ hosts; in this example, $H = 4$, which would result in 14 hosts.
3. $S + H <= 8$ (Class C network); in this example $4 + 4$ is less than or equal to 8.

- S+H =< 8 4+4=<8 valid

Step 3

- Determine the subnet mask (S=4)
- 4bits in 4th octane 128 + 64 + 32 + 16 = 240.
- Step-4 Determine network address

192.168.1.0	192.168.1.64	192.168.1.128	192.168.1.192
192.168.1.16	192.168.1.80	192.168.1.144	192.168.1.208
192.168.1.32	192.168.1.96	192.168.1.160	192.168.1.224
192.168.1.48	192.168.1.112	192.168.1.176	192.168.1.240

Step -5

- Determine broadcast address

Network Address	Mathematics	Directed Broadcast Address
192.168.1.0	16 – 1	192.168.1.15
192.168.1.16	32 – 1	192.168.1.31
192.168.1.32	48 – 1	192.168.1.47
192.168.1.48	64 – 1	192.168.1.63
192.168.1.64	80 – 1	192.168.1.79
192.168.1.80	96 – 1	192.168.1.95
192.168.1.96	112 – 1	192.168.1.111
192.168.1.112	128 – 1	192.168.1.127
192.168.1.128	144 – 1	192.168.1.143
192.168.1.144	160 – 1	192.168.1.159
192.168.1.160	176 – 1	192.168.1.175
192.168.1.176	192 – 1	192.168.1.191
192.168.1.192	208 – 1	192.168.1.207
192.168.1.208	224 – 1	192.168.1.223
192.168.1.224	240 – 1	192.168.1.239
192.168.1.240		192.168.1.255

Step 6: Determine the Host Addresses

Network Number	Host Address	Directed Broadcast Address
192.168.1.0	192.168.1.1–192.168.1.14	192.168.1.15
192.168.1.16	192.168.1.17–192.168.1.30	192.168.1.31
192.168.1.32	192.168.1.33–192.168.1.46	192.168.1.47
192.168.1.48	192.168.1.49–192.168.1.62	192.168.1.63
192.168.1.64	192.168.1.65–192.168.1.78	192.168.1.79
192.168.1.80	192.168.1.81–192.168.1.94	192.168.1.95
192.168.1.96	192.168.1.97–192.168.1.110	192.168.1.111
192.168.1.112	192.168.1.113–192.168.1.126	192.168.1.127
192.168.1.128	192.168.1.129–192.168.1.142	192.168.1.143
192.168.1.144	192.168.1.145–192.168.1.158	192.168.1.159
192.168.1.160	192.168.1.161–192.168.1.174	192.168.1.175
192.168.1.176	192.168.1.177–192.168.1.190	192.168.1.191
192.168.1.192	192.168.1.193–192.168.1.206	192.168.1.207
192.168.1.208	192.168.1.209–192.168.1.222	192.168.1.123
192.168.1.224	192.168.1.225–192.168.1.238	192.168.1.239
192.168.1.240	192.168.1.241–192.168.1.254	192.168.1.255

Determining IP Address Components

- When you are given a particular address and subnet mask, and asked whether the address is a network, host, or directed broadcast address, you should use the following six steps:

- 1. You need an IP address and a subnet mask (this is the easy part).
- 2. Examine the subnet mask and find the interesting octet. The interesting octet in the mask is the one in which the network and host boundary are found.
- 3. Subtract the interesting octet in the subnet mask from 256. This will give you the increment by which network numbers are increasing in the interesting octet.

- 4. On a piece of paper, start writing down the network numbers, starting with the first subnet (0), and working your way up to a network number that is higher than the address in question.
- 5. After you have written down the network numbers, beside each of these, write down their corresponding broadcast addresses.
- 6. Between the network and broadcast addresses, write down the host addresses. Host addresses are any number between the network and directed broadcast addresses.

Example #1

- **172.16.5.0 255.255.254.0** or 172.16.5.0/23
- network and host boundary resides: 3rd
- 256 – 254 = 2
- *Network numbers are incrementing by 2 in the third octet*
- 172.16.0.0,172.16.2.0, 172.16.4.0, 172.16.6.0, 172.16.8.0, and so on and so forth
-

Class B

Network Address	Host Address	Directed Broadcast Address
172.16.0.0	172.16.0.1–172.16.1.254	172.16.1.255
172.16.2.0	172.16.2.1–172.16.3.254	172.16.3.255
172.16.4.0	172.16.4.1–172.16.5.254	172.16.5.255
172.16.6.0	172.16.6.1–172.16.7.254	172.16.7.255
172.16.8.0	172.16.8.1–172.16.9.254	172.16.9.255